

Microscopic reversibility as the organizing principle of molecular machines

R. Dean Astumian

Biological motors and pumps are equilibrium devices that couple chemical, electrical and mechanical processes in an environment that is far from equilibrium. Recognition of the key role played by microscopic reversibility in their operation is a first step towards rational design of artificial molecular devices.

Over 50 years ago an engineer, William McClellan, constructed a motor that easily fit into a cube 1/64th of an inch (about 4×10^{-4} m) on a side in response to a challenge made by Richard Feynman. The tiny unpowered motor is now displayed at the Cal Tech museum. When viewed under a microscope, the motor does exactly what any other macroscopic motor does in the absence of input power — nothing. The organizing principle for McClellan's machine, and for all macroscopic machines, even very small ones, is cause and effect — the machine is constructed such that an applied force is transmitted through the device to cause a desired output. Unless

a force is applied the motor remains in static equilibrium.

To understand the very significant differences between the physical principles that govern the operation of even very small macroscopic machines such as McClellan's motor and molecular machines consider a molecular 'walker' shown in Fig. 1. The molecular walker is inspired by the biomolecular motor kinesin, an important component of protein-based nanotechnology applications. This walker uses chemical energy provided by the hydrolysis of adenosine triphosphate (ATP) to step in one direction along a macromolecular track (Fig. 1). Even at chemical equilibrium where the

chemical potential of ATP is equal to that of adenosine diphosphate (ADP) plus inorganic phosphate (P_i), the molecular motor, because of thermal noise, continually steps back and forth along its track and continually catalyses back and forth chemical conversions between ATP, and ADP plus P_i . As with all molecular systems, equilibrium of a molecular machine is a dynamic state in which all motions, including those that are used for the function of the machine, are present, but where every forward motion is exactly cancelled by the microscopic reverse of that motion.

The schematic in Fig. 1 gives rise to a simple, mechanically based, narrative. First, ATP binds and causes dissociation of the rear head from the track. Stored elastic energy propels the head forward in a 'power stroke' to a position near the next binding site. Following hydrolysis of ATP and dissociation of ADP and P_i , the head reattaches, predominately to this nearest forward-binding site. Continuing the narrative we ask what is it that ATP 'does' to drive directed motion? The natural answer is that ATP 'deposits energy', sometimes signified by a 'flash' in animations of molecular machine movement. The subsequent 'non-equilibrium vibrational energy relaxation' ultimately drives the motion by inducing the long-range conformational change called the power stroke.

When we step back, however, and critically analyse this mechanical narrative we recognize that the story makes no sense from the perspective of a single molecule in water. Changes in configuration of any macromolecule after binding and/or hydrolysis of ATP must on average

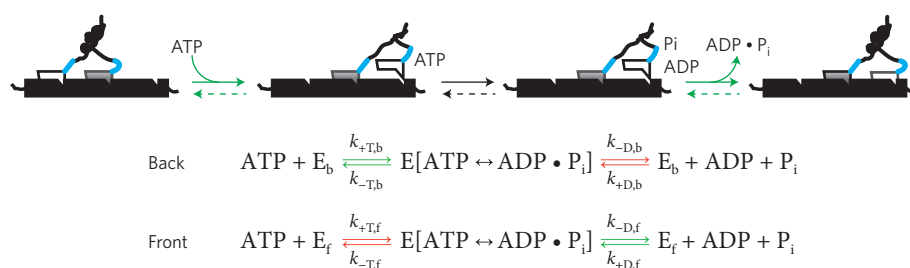


Figure 1 | Illustration of an ATP hydrolysis powered molecular walker. Top: schematic of the mechanism for a molecular walker with two *a priori* identical heads, one white and the other grey moving on a macromolecular track. When both heads are bound to the polar track, whichever head is bound in front is chemically distinct from the head that is bound at the back. The sequence of steps from left to right describes a process in which the walker takes one step forward (to the right) and one ATP is hydrolysed to ADP and P_i . Bottom: the corresponding Michaelis-Menten equations for both the back (E_b) and front (E_f) heads are shown. The key ingredient allowing directional motion is the specificity difference for a head bound in the back position and one bound in the front position. This is illustrated by the green arrows that indicate fast reaction rates for the binding and release of ATP at the back head and for the binding and release of ADP and P_i at the front head, and the red arrows that indicate slow rates for the binding and release of ADP and P_i at the back head and release of ATP at the front head. The black double headed arrow ($\leftrightarrow$) between ATP, and ADP $\cdot P_i$ indicates that the interconversion between these forms at the active site is freely reversible in the bound form of each catalytic head.

be exactly the same both in solutions where ATP in the bulk is not in equilibrium with ADP and P_i and in solution where they are in equilibrium. There is no mechanism by which the molecular machine can sense the bulk concentrations of ATP, ADP and P_i . Hence, **the so-called power-stroke is an equilibrium process. Therefore, movies of an individual molecular machine following ATP binding until reattachment of the head to the track must be statistically identical both in systems where ATP in the bulk is far from thermodynamic equilibrium and in systems where ATP is at thermodynamic equilibrium with ADP and P_i .** The only difference between a situation with ATP away from rather than at chemical equilibrium is that **the relative probabilities to bind ATP versus ADP are shifted** — the effect of ATP is based on mass action.

Instead of the mechanical picture let us **take a chemists view of the molecular walker**. Each head, back (E_b) and front (E_f), acts as an enzyme (E) to catalyse the reaction between ATP and ADP + P_i . The two mechanisms can be expressed through the Michaelis–Menten equations reported in Fig. 1.

In terms of the rate constants for the Michaelis–Menten mechanisms shown in Fig. 1, the overall equilibrium constant for the catalysed reaction ($ATP \rightleftharpoons ADP + P_i$) is $k_{+T,i}k_{-D,i}/(k_{-T,i}k_{+D,i})$, where k_T and k_D are the dissociation constants for ATP and ADP respectively, and $i = f, b$. This equilibrium constant is of course the same at the front (f) and back (b). Further, because the bound form of the enzyme $E[ATP \leftrightarrow ADP \cdot P_i]$ is a common intermediate for the reactions involving the front head and the back head, the dissociation constants given by the ratios of the on and off rate constants for ATP and ADP, $k_{+T,i}/k_{-T,i}$ and $k_{+D,i}/k_{-D,i}$, respectively, must be the same at the front and back. **These relationships between the rate constants express the thermodynamically required constraints of the principle of microscopic reversibility for the mechanism. What need not be the same at the front and back are the relative specificities (that is, the ratios of the ‘off’ rates for ATP and ADP), $s_i = k_{-b,i}/k_{-D,i}$. If $s_b/s_f \gg 1$ the binding/release of ATP will be much faster than binding/release of ADP at the back head, and the binding release of ADP will be much faster than the binding/release of ATP at the front head. For the molecular walker shown in Fig. 1, the green and red arrows indicate fast and slow processes, respectively. A likely mechanism for the specificity switching involves allosteric feedback, possibly mediated by strain, between the neck linker (blue region in Fig. 1) and the ATPase active site². Because of the differences in the rates,**

Table 1 | Comparison between macroscopic and molecular machines.

Macroscopic mechanical devices	Microscopic chemical devices
Cause and effect	Microscopic reversibility
No motion without input energy	Incessant thermal motion
Far from mechanical equilibrium	Very near to mechanical equilibrium
Applying an external force causes forward motion	Turning on energy barriers prevents backward motion
Statistical correlations unimportant	Statistical correlations essential

the most probable mechanism for catalysis is the one that follows the green arrows, $ATP + E_b \rightleftharpoons E[ATP \leftrightarrow ADP \cdot P_i] \rightleftharpoons E_f + ADP + P_i$ where the heads switch positions as shown in Fig. 1. **Under physiological concentrations, where ATP is in great excess of its equilibrium concentration, the reaction is shifted by mass action strongly in the direction favouring ATP hydrolysis and stepping forward.** It should be noted the motor moves to the rear if $s_b/s_f < 1$. The load-free direction of motion depends only on the ratio of the specificities and not on the structural details of any power stroke.

Under a sufficiently strong force to the rear the motor can, in principle, step backward by reversing the most probable mechanism and thereby use mechanical energy to synthesize ATP against the strong thermodynamic tendency to hydrolyse ATP. Alternatively, **the load may instead overcome the unfavourable kinetic specificities and hydrolyse ATP while taking a step backward** by the mechanism $ATP + E_f \rightleftharpoons E[ATP \leftrightarrow ADP \cdot P_i] \rightleftharpoons E_b + ADP + P_i$. This latter possibility was predicted³ in 1996 and supported in 2005 by experiments on kinesin suggesting that back-stepping under a strong load is stimulated rather than inhibited by ATP⁴. **The thermodynamic properties of the motor** — the ratio of forward to backward steps, the thermodynamic efficiency and the force necessary to cause the motor to move backward — **are governed by the specificities⁵. The power stroke**, that is, the free energy released in bringing the back head forward on detachment from the track, **is important in determining the kinetic properties of the motor — the velocity, the power output and the force–velocity curve.**

An individual molecular machine can be viewed as **a mechanically equilibrated system that serves as a conduit for exchange of energy between chemical and physical reservoirs that may be far from thermodynamic equilibrium** with one another. I argue here that molecular machines in solution operate as Brownian motors⁶ in which the key mechanism involves prevention of undesired motion (for example, no binding of ATP at the front

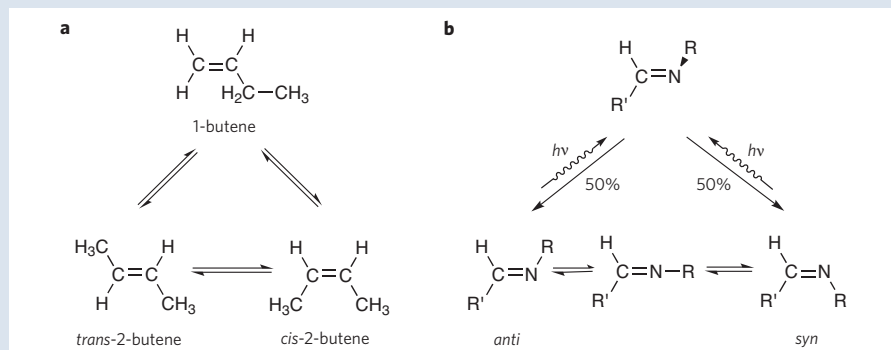
head or release of ADP at the back head) rather than causing desired motion, and that **the organizing principle by which molecular machines can be understood is microscopic reversibility⁷.** This understanding will play a strong role in the design, characterization and operation of molecular machines for use in nanotechnology. Table 1 lists several characteristics of macroscopic machines that are compared with the corresponding characteristics of molecular machines.

Microscopic reversibility and binding

A good way to understand microscopic reversibility at work is to consider the binding of carbon monoxide to myoglobin. After carbon monoxide binds to the heme group of myoglobin the iron moves into the plane of the heme, and then a long-range conformational change propagates away from the heme. It is natural to hypothesize in the reverse de-binding process that the heme group first returns to the out-of-plane configuration and then the global conformational change occurs to restore the protein to its original unbound structure. This common-sense order of local change followed by global change in the de-binding process is supported by experiments⁸ of Frauenfelder and colleagues in which carbon monoxide is photochemically dissociated from myoglobin. The results of the photo-dissociation experiments led Ansari *et al.*⁸ to conclude, “binding or dissociation of a ligand from the heme iron causes a protein quake”. This conclusion, however, when applied to thermal processes is not consistent with the principle of microscopic reversibility by which the mechanism for a reaction is independent of its direction (Box 1). Counterintuitive though it may be, thermal dissociation of a ligand, even from a large complex protein, must, on average, occur by the microscopic reverse of the binding process of that ligand — that is, thermal dissociation of carbon monoxide must, on average, be preceded by an ‘un-quake’, followed by out-of-plane repositioning of the iron centre and release of carbon monoxide.

The constraints of microscopic reversibility can be circumvented if the

Box 1 | Microscopic reversibility.



In a reversible reaction, the mechanism in one direction is exactly the reverse of the mechanism in the other direction. This does not apply to reactions that begin with a photochemical excitation²².

A chemical example illustrating the implications of microscopic reversibility is the isomerization of butene describable by a triangle reaction. As discussed by Onsager⁷ each forward and backward process in (a) must balance independently by microscopic reversibility — there is no possibility for net cycling.

Lehn has given a very interesting conjectural example of how photochemistry can be used to cause an imine to act as a rudimentary molecular rotor²³ shown in (b), where steric interactions favours the *anti*- form. In the ground state the imine must be planar and thermal conversion between the *syn*- and *anti*- forms occurs via a linear intermediate. When excited, the R group

moves out of plane. If the molecule is immobilized on a surface the only possibility is for the R group to be above the plane. De-excitation occurs with equal probability to the *syn*- and *anti*- form, followed by thermal re-equilibration to the original unbalanced distribution between *syn* and *anti*. The overall effect is net counterclockwise rotation as seen from looking down the axis from N to C. Such cyclic motion by a molecule immobilized on a surface would result in pumping liquid²⁴.

A similar mechanism for continual cyclic rotation based on chemical driving has been demonstrated experimentally by Feringa and colleagues²⁵. Their approach uses a chiral ring-opening reagent. Successive ring openings and closings, enforced by external changes in the solution, leads to directed rotation. The direction of rotation can be reversed by choosing the opposite chirality for the reagent.

macromolecule is a catalyst because the ligand that binds (substrate) can be different than the ligand that is released (product). In a cycle of catalytic turnover, the conformational pathway following release of product is not in general the microscopic reverse of the conformational pathway following binding of a molecule of substrate. The turnover allows repetitive cycling, a key difference between machines and simple switches⁹. The catalytically induced conformational cycling is illustrated by recent experiments of Sen and colleagues¹⁰ showing that the apparent diffusion constant of a 'garden-variety' enzyme, urease, in solution is significantly increased when it is catalytically active. An interpretation of this observation is that each catalytic turnover moves the enzyme a certain distance because of the cyclic conformational change, but the *a priori* directed motion seems random (diffusive)

because the enzymes in solution are randomly oriented. These results suggest that many enzymes could act as molecular machines to harness chemical energy to do work if they could be oriented. Chemists have recently made great progress towards this goal by adapting nature's trick of orienting molecular catalysts on a track coupled with an allosteric interaction by which the chemical activities become position dependent.

Nature's own Maxwell's demon

The direction of net motion of a molecule diffusing in one dimension on a polymeric track depends solely on the direction of any applied force. Now, **imagine an intelligent being like a Maxwell's demon that can observe the molecule and place obstacles behind, but not in front of it.** If the obstacles are insurmountable, **the molecule moves inexorably forward**, albeit possibly

very slowly, even against a strong opposing force. **The demon placing the obstacles does no 'force times distance' work on the diffusing molecule.** Instead, **information about the molecules position is used by the demon to determine where to place the obstacles.** For this reason, the mechanism is called an **information ratchet**¹¹.

David Leigh and colleagues have implemented such an information ratchet chemically¹². They start with a 2-rotaxane, a supramolecular structure with a linear polymer that serves as the track threaded through a macromolecular ring. The ends of the polymer have very bulky groups to prevent the ring from slipping off into solution. There are two *a priori* identical interaction stations on the track where the ring can be localized, and a hydroxyl group separating the stations. Because the hydroxyl group is very small, the ring undergoes rapid diffusion back and forth between the two stations. Benzoylation of this hydroxyl with an achiral catalyst replaces the hydroxyl group with a bulky benzoyl group that prevents exchange between the two sites. Nuclear magnetic resonance analysis shows that after achiral benzoylation the ring is equally apportioned between the two stations. When a chiral benzoylation catalyst is used, however, the resulting apportionment between the two stations is not symmetric. The entropy of the system is decreased by benzoylation with a chiral but not an achiral catalyst. Further, if different interaction stations with different interaction energies are used, a chiral benzoylation catalyst can disproportionately populate the higher-energy station, thereby doing enthalpic work. This mechanism is possible because the position of the ring influences the rate of the catalytic benzoylation by allosteric feedback. Let us consider a similar mechanism where the ring carries the catalyst as its own portable power generator¹³, and where the track is many periodic units long to illustrate how allosteric feedback provides a general mechanism for autonomous conversion of chemical energy into mechanical work with a high efficiency.

Consider a ring diffusing on a periodic track consisting of two distinct subunits or bases (A,B) as shown in Fig. 2a with gates to the left of each A that can be either open as in the top row, allowing free diffusion, or closed as in the bottom row, preventing diffusion. The subscripts $-1, 0, 1, \dots i$ indicate the position along the track. One chemical approach to implement a demon is to add a catalyst that interconverts some substrate–product pair, S and P, on the diffusing ring (Fig. 2a) such that when the

active site is occupied (indicated by the filled inner blue ring, or the superscript asterisk on the state labels) the repulsive interaction that prevents diffusion across the barriers (grey balls) is quenched. If allosteric interactions can be designed that allow the catalyst to be specific for P when the ring is on base A and specific for S when the ring is on base B, away from chemical equilibrium with $[S]K_{sp}/[P] > 1$ the gates will more likely to be open when the ring is on base B and more likely to be closed when the ring is on base A (Fig. 2b). The factor K_{sp} is the equilibrium constant for the catalysed reaction. The correlation between the position of the ring and the likelihood for the gates to be open or closed away from equilibrium is sufficient to induce directed motion¹¹ to the right, even against an applied force. When the reaction is at equilibrium, however, the gates are equally likely to be up or down irrespective of where the ring is and, consistent with the second law, there is no directed motion.

The information ratchet works by preventing backward motion — when the ring moves onto base A product most probably rapidly dissociates thereby closing the gates to prevent backward steps. The forward, desired motion occurs by thermal noise after the substrate binds when the ring is on station B. Irrespective of the details of the mechanism, the key to obtaining directed motion is to engineer a molecule with large allosteric interactions so that the specificities for binding S and P on bases A and B are very different.

Allosteric interaction provides an explicit mechanism for implementation of a Maxwell's demon by a molecular machine, but only when the driving chemical reaction is away from equilibrium.

Allosteric interaction provides an explicit mechanism for implementation of a Maxwell's demon by a molecular machine, but only when the driving chemical reaction is away from equilibrium.

Many chemical fuel reactions, such as, hydrogenation, oxygenation, or reduction/oxidation processes, provide so much chemical energy (more than 50 to 60 times the thermal energy per conversion, which is 2 to 3 times the energy provided by ATP hydrolysis) that these reactions are effectively irreversible. If the free energy released by the reaction $S \rightleftharpoons P$ is

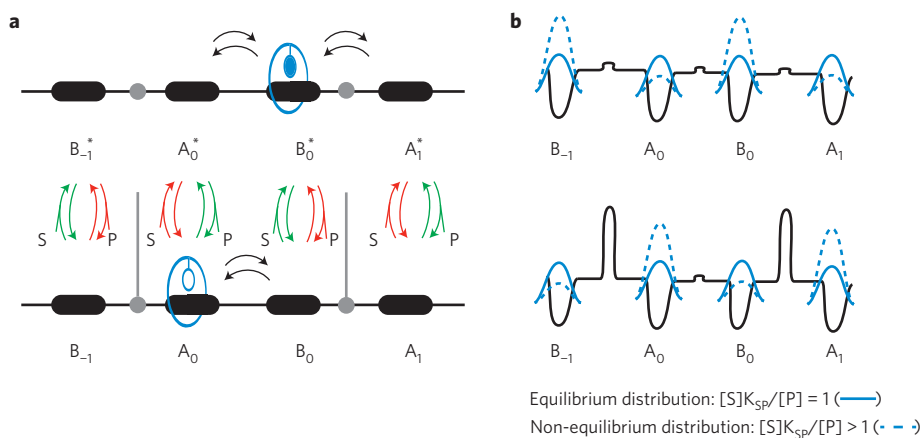


Figure 2 | An information ratchet that uses allosteric interactions as a mechanism for feedback between the opening and closing of gates and the position of a ring on a polymeric track. **a**, A ring with a catalyst (blue) diffuses on a linear polymeric track. When the catalytic active site is unoccupied (bottom row; inner open blue circle) a strong repulsive interaction prevents diffusion across the grey site to the left of every A. Binding of S or P to the active site, indicated by an asterisk, quenches the repulsive interaction, allowing free diffusion. The binding processes are indicated by the curved red and green arrows leading from the bottom to the top row, and the dissociation processes are indicated by the curved red and green arrows leading from the top to the bottom row. The ratios of all equilibrium constants between the states are constrained by microscopic reversibility. The ratios of the off rates for S and P — the specificities — are not constrained by microscopic reversibility. In general, the chemical environment for a ring with the grey interaction site to the left is different than that for a ring with the grey interaction site to the right, and hence the specificities for S and P are different on the sites A and B. This environment-dependent kinetics is known as an allosteric interaction because it can arise from effects away from the active site. Consider a catalytic cycle that starts at A_0 . If the concentration of P is low, the ring will most probably next move to B_0 . At B_0 , if the concentration of S is high the catalyst quickly binds to it thereby arriving in state B_0^* ; the complex moves to an A^* station to the back (A_0^*) or front (A_1^*). In either case the ring will most probably dissociate P, returning to state A_0 , or completing a forward step to A_1 . **b**, Schematic of the potential energy profile for the bound (upper) and unbound (lower) states of the motor (black line). Distribution of ring positions is shown schematically for the equilibrium case (solid blue curve) and the non-equilibrium case (dashed blue curve). In the equilibrium case, the probability to find the ring in either A or B, bound or unbound states is the same; however, because of the specificity differences in the non-equilibrium case, the ring is most likely to be found on station B when the gates are open and most likely on station A when the gates are closed, giving rise to directed motion to the right.

very large for the information ratchet in Fig. 2, reversal of the forward process $A_0 \rightleftharpoons B_0 \rightleftharpoons B_0^* \rightleftharpoons A_1^* \rightleftharpoons A_1$ that couples forward stepping to catalysis of S to P is very unlikely, even in the presence of a strong backward directed force. However, the alternative backward process $A_0 \rightleftharpoons A_0^* \rightleftharpoons A_0^* \rightleftharpoons B_{-1}^* \rightleftharpoons B_{-1} \rightleftharpoons A_{-1}$ couples backward stepping to catalytic conversion of S to P if the force is strong.

Challenges for the future

A traditional picture of macromolecular dynamics involves 'non-equilibrium' functionally important motions⁸ — power strokes, protein quakes and so on — that are separate from equilibrium conformational fluctuations. Thus it has been argued that the description of molecular machines that convert chemical energy to mechanical work "must be done in the language of mechanics: 'moving

parts', forces, torques, displacements, thermodynamic efficiencies and time."¹⁴ A logical expectation based on this perspective is that a 'movie' of a molecular machine carrying out its function, based on many X-ray structures obtained under many conditions should, by revealing the functional motion, provide a more or less complete understanding of how that machine works and how it can be adapted for nanotechnology applications. Consideration of microscopic reversibility, however, challenges this perspective.

Unlike photochemical processes, binding and thermal dissociation of any single ligand (including ATP) occur most probably by the microscopic reverses of one another. Because the physical motions of the macromolecule following ligand binding, catalytic conversion, and dissociation of ligand are all equilibrium processes¹⁵ irrespective of the bulk chemical potentials

of the ligands, there is no dissipative loss due to internal friction and no fundamental limit to the efficiency of molecular machines lower than 100%. The movie that has served as the holy grail of structural biology is nothing more than a series of snapshots of equilibrium structures with the sense of time inserted by hand based on knowledge of the sequence of the chemical events of interest, and where the motion is smoothed by interpolation between the discrete states¹⁶.

Paradoxically, the key steps that determine the function and directionality of a molecular machine driven by a catalysed chemical reaction are those in which the molecular machine itself moves little if at all — that is, those branch points in the catalytic cycle where the machine binds or de-binds the substrate or product of the driving reaction¹⁷. The selection process is controlled by allosteric interactions between the position or state of the machine and its catalytic active site. Coskun *et al.*⁹ have suggested that coupling two separate redox reactions to a simple switch could transform the device into an autonomous molecular machine by allowing continuous catalysis of the energy releasing process.

Viewing molecular machines in their proper context as first and foremost molecules that operate by the laws of chemistry, rather than as miniaturized macroscopic devices, is beginning to pay dividends in nanotechnology. Recent work has focused on chemical control of switching lability¹⁸ and stability of molecular states¹⁹, and on controlling the chiral space of a catalyst²⁰. Insights from this chemical perspective may prove critical in designing molecular machines that act as ‘mechanized’ enzyme-like devices that can use input chemical energy to assemble a wide range of novel, high-energy, ordered materials²¹. □

R. Dean Astumian is at the Department of Physics, The University of Maine, 5709, Bennett Hall, Orono, Maine 04469-5709, USA.
e-mail: astumian@maine.edu

References

1. Von Delius, M. & Leigh, D. A. *Chem. Soc. Rev.* **40**, 3656–3676 (2011).
2. Hyeon, C. B. & Onuchic, J. *Biophys. J.* **101**, 2749–2759 (2011).
3. Astumian, R. D. & Bier, M. *Biophys. J.* **70**, 637–653 (1996).
4. Carter, N. J. & Cross, R. A. *Nature* **435**, 308–312 (2005).
5. Astumian, R. D. *Biophys. J.* **98**, 2401–2409 (2010).
6. Astumian, R. D. & Hanggi, P. *Phys. Today* **55**, 33–39 (November 2002).

7. Onsager, L. *Phys. Rev.* **37**, 405–426 (1931).
8. Ansari, A. *et al. Proc. Natl Acad. Sci. USA* **82**, 5000–5004 (1985).
9. Coskun, A., Banaszak, M., Astumian, R. D., Gryzbowski, B. & Stoddart, F. *Chem. Soc. Rev.* **41**, 19–30 (2012).
10. Muddana, H. S., Sengupta, S., Mallouk, T. E., Sen, A. & Butler, P. J. *J. Am. Chem. Soc.* **132**, 2110–2111 (2010).
11. Astumian, R. D. & Derenyi, I. *Eur. Biophys. J.* **27**, 474–489 (1998).
12. Alvarez-Perez, M., Goldup, S. M., Leigh, D. A. & Slawin, A. M. Z. *J. Am. Chem. Soc.* **130**, 1836–38 (2008).
13. Thordarson, P., Bjsterveld, E. J. A., Rowan, A. E. & Nolte, R. J. M. *Nature* **424**, 915–918 (2003).
14. Bustamante, C., Cheng, W. & Mejia, Y. X. *Cell* **144**, 480–497 (2011).
15. Astumian, R. D. *Am. J. Phys.* **74**, 683–688 (2006).
16. Moore, P. *Ann. Rev. Biophys.* **41**, 1–19 (2012).
17. Astumian, R. D. *Ann. Rev. Biophys.* **40**, 289–313 (2011).
18. Share, A. I., Parimal, K. & Flood, A. H. *J. Am. Chem. Soc.* **132**, 1665–1675 (2010).
19. Kay, E. R., Leigh, D. A. & Zerbetto, F. *Angew. Chem. Int. Ed.* **46**, 72–191 (2007).
20. Wang, J. & Feringa, B. L. *Science* **331**, 1439–1441 (2011).
21. Sengupta, S., Ibele, M. E. & Sen, A. *Angew. Chem. Int. Ed.* **51**, 8434–8445 (2012).
22. International Union of Pure and Applied Chemists (IUPAC) Compendium of Chemical Terminology; <http://goldbook.iupac.org>
23. Lehn, J. M. *Chem. Eur. J.* **12**, 5910–5915 (2006).
24. Purcell, E. *Am. J. Phys.* **45**, 3–11 (1977).
25. Fletcher, S. P., Dumur, F., Pollard, M. M. & Feringa, B. L. *Science* **310**, 80–82 (2005).

Acknowledgements

I am grateful for very useful correspondence with Imre Derenyi, Chang-Bong Hyeon, Peter Moore, Dwayne Miller, Ayusman Sen, Sean Sun, and for useful discussions with Nicki Curthoys.